Learning to Recognise Gravitationally Lensed Fast Radio Bursts Without Real Positives

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***Abstract*—A fast radio burst (FRB) lensed by a foreground galaxy reaches Earth twice, days to months apart, and the two copies differ only by propagation effects. Recognising such copies among the bursts of repeating sources is a pairwise verification problem with three difficulties: no real positive example exists, the negatives include adversarial *impostors* that share a propagation imprint, and decisions must hold at a 1% false-positive rate. We build a semi-synthetic benchmark from all 981 public dynamic spectra of the 83 repeating sources in the CHIME/FRB Catalog 2, and propose a 259k-parameter Siamese network that aligns the two bursts' feature maps with a differentiable correlation layer before fusing them. In a source-grouped cross-validation rehearsal on the 60 development sources it recovers 0.47 [0.25, 0.66] of lensed pairs at 1% false-positive rate, against 0.25 [0.20, 0.30] for the best physics- or feature-based baseline within the simulated nuisance ranges, and ablations over three seeds show that both alignment and impostor training are needed. Because the positive class has no real counterpart, we validate on real data in three ways: real cross-source pairs serve as true negatives scored against per-pair simulated twins (false-positive rate 0.8–1.5%, within a pre-registered 2% limit), two copies of a real burst stand in for a lensed pair (recall ratio 0.96–0.97 to template copies), and the whole pipeline is reproduced on a second machine. Finally, we show that survey geometry, not the classifier, limits discovery: a transit telescope records both copies for only 1.8% of lenses, and an all-sky monitor would find about $\sim$50$\times$ more lensed repeaters from the same sources.**

***Index Terms*—Siamese networks, pairwise verification, simulation-to-real transfer, deep learning for signal processing, radio astronomy**

I. Introduction

Fast radio bursts are millisecond radio flashes from distant galaxies [1], [2], and some sources repeat thousands of times. When a galaxy lies close to the line of sight, gravitational lensing forms two images of the source, so every burst arrives twice, separated by a fixed delay Δt of days to months (Fig. 1a). A confirmed lensed FRB would measure the expansion rate of the Universe and test dark-matter models [3]–[5], but none has been found [6]–[9].

Finding one is a pairwise verification problem: are two bursts from the same sky position two copies of one burst, or two different bursts of a repeating source? Three properties make it hard. *No real positives:* any detector is trained on simulation, and its transfer to real data must be checked without labelled positives. *Adversarial negatives:* distinct bursts of one source can share a spectral envelope imposed by a plasma lens, or a scintillation pattern, and resemble copies. *Low operating point:* the probability that a given repeater is strongly lensed is 10^{-4} or less (the lensing optical depth is 1.5×10^{-4} even at redshift 1), so decisions must hold at 1% false-positive rate or below.

This paper makes four contributions (Fig. 2):

1) a semi-synthetic pair benchmark built from all public CHIME/FRB Catalog 2 repeater spectra, with impostor negatives and frozen, hash-checked evaluation sets;
2) a Siamese correlation network that aligns feature maps before comparing them, with an ablation study over three seeds;
3) a validation protocol for a detector without real positives, combining true-negative transfer with per-pair twins, a real-burst positive proxy and an independent reproduction;
4) an analysis of where such a detector can work, showing that transit telescopes rarely record both copies and that sky coverage, not classifier accuracy, limits discovery.

II. Related Work

Siamese verification. Siamese networks decide whether two inputs share an identity by comparing the outputs of one shared encoder, from signature verification [10] and contrastive embedding [11] to one-shot recognition [12]. Most compare global embedding vectors, which discards where features occur; patch-matching studies contrast such networks with two-channel networks that fuse the inputs early [13], and our baselines include both. Comparing feature maps over a range of displacements was introduced for optical flow [14] and used for Siamese tracking [15]; we use it to estimate and remove the residual time offset between two bursts before comparing them.

Lensed-pair identification. For gravitational waves, lensed event pairs have been identified with a DenseNet and gradient boosting [16], and with a Siamese residual network that fuses features by absolute difference and product [17], which notes the need to adapt to real detector data. FRB lensing searches have used auto- and cross-correlation statistics [6], [7], [9], and machine learning on FRBs has separated repeating from non-repeating sources [18]; to our knowledge no learned detector of lensed FRB copies exists.

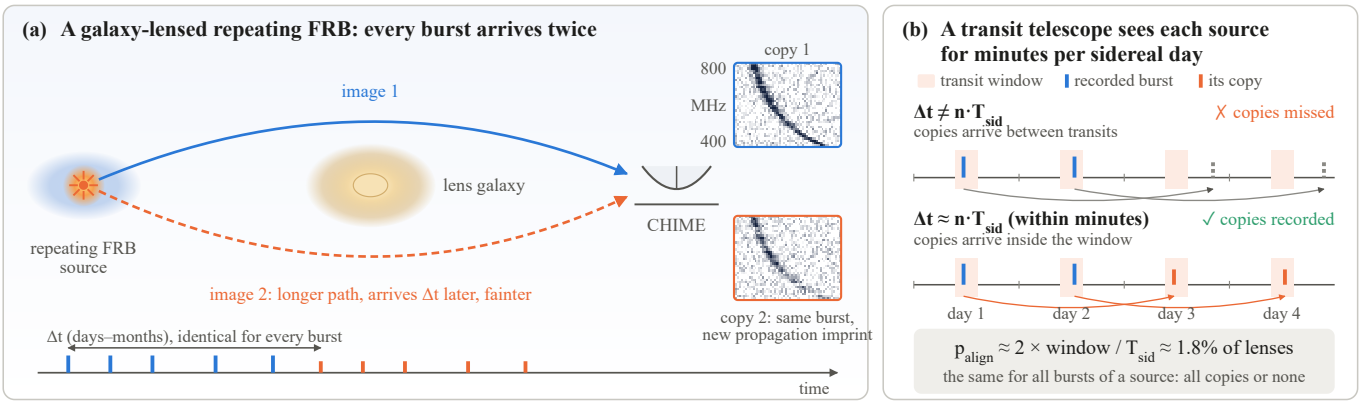


Fig. 1. The problem. (a) A galaxy lens sends every burst of a repeating FRB along two paths; the copy arrives a fixed Δt later, fainter, with its own scintillation and scattering. The dynamic spectra show one burst sweeping from 800 to 400 MHz (the ν^{-2} dispersion delay), computed from the cold-plasma law. (b) A transit telescope such as CHIME records a source only within minutes of transit, so the copies of recorded bursts are recorded only if Δt is close to a whole number of sidereal days.

Simulation-to-real transfer. Detectors trained on simulation are made robust by randomising nuisances during training [19], but robustness still has to be shown on real data. Without real positives, we check transfer on real negatives, per-pair simulated twins and a real-signal positive proxy.

III. Problem Formulation

A burst is a dynamic spectrum $X \in \mathbb{R}^{F\times T}$ with $F = 256$ frequency channels and $T = 160$ time samples of 0.983 ms, normalised to unit off-pulse noise. A pair (A, B) of bursts from one sky position has a label $y \in \{\mathrm{L}, \mathrm{R}, \mathrm{U}, \mathrm{I}\}$: lensed copies, repeat (two bursts of one source), unrelated (two sources) or impostor (two bursts of one source sharing a propagation imprint). A detector outputs a symmetric score $s(A, B) = s(B, A)$, and a pair is flagged if $s > \tau_\alpha$, where τ_α is the $(1-\alpha)$ quantile of s over negatives. The primary metric is

$$\mathrm{Recall@}\alpha = \Pr[s(A, B) > \tau_\alpha \mid y = \mathrm{L}], \quad \alpha = 1\%, \qquad (1)$$

with negatives $\{\mathrm{R}, \mathrm{I}\}$, the alternatives that a real search faces within one source. Pairs share sources, so we report 95% intervals from a bootstrap over sources that recomputes τ_α in every resample. The validation constraint is that no real pair with $y = \mathrm{L}$ is available: real data can only supply negatives.

IV. Benchmark Construction

Real inputs. We use the Stokes-I dynamic spectra of all 981 bursts of the 83 repeating sources in CHIME/FRB Catalog 2 [20], recorded by the CHIME transit telescope [21] at 400–800 MHz. Each burst is binned to 256×160 around its emission and normalised by its off-pulse noise. The catalogue's fitted burst models serve as noise-free templates (950 usable), and the off-pulse columns of each file, mirror-tiled, serve as real noise backgrounds (856), each with its own channel mask.

Pair classes. A lensed pair places one template twice; a repeat pair, two templates of one source; an unrelated pair, templates of two sources. An impostor pair is a repeat pair in which member B inherits member A's 25-MHz-smoothed spectral envelope (I1, mimicking a plasma lens), both members share one scintillation pattern (I2), or both. Each member is injected at a measured matched-filter signal-to-noise ratio (S/N) on a different real background, with its own time shift (± 10 samples), residual dispersion, scattering and scintillation, so that only burst identity separates lensed from repeat pairs. Probes on single-member statistics cannot tell the classes apart (AUC ≤ 0.53).

Splits and frozen sets. Sources are split 60/20/20 into train, validation and test. Validation and test sets of 4000 pairs, and "bright" sets with pair minimum S/N ≥ 25 (the pre-registered headline bin), are regenerated from fixed seeds and checked against stored SHA-256 hashes. The test split stays sealed under a pre-registered blind protocol; all numbers below are development results on sources never used in training.

V. Siamese Correlation Network

Encoder. Inputs pass through asinh to compress the dynamic range of bright bursts. A shared encoder g of four 5×3 convolution blocks with group normalisation maps X to $g(X) \in \mathbb{R}^{C\times F'\times T'}$ with $C = 64$, $F' = 16$ and $T' = 80$, keeping two-sample time resolution because lensed copies differ by residual time offsets.

Correlation and soft alignment. For lags $\ell \in \{-L, \ldots, L\}$ with $L = 12$ encoder steps (± 24 samples),

$$\rho(\ell) = \frac{\langle \bar{g}_A,\ \bar{g}_B(\ell)\rangle}{\|\bar{g}_A\|\,\|\bar{g}_B(\ell)\|}, \qquad w_\ell = \frac{e^{\beta\rho(\ell)}}{\sum_k e^{\beta\rho(k)}}, \qquad (2)$$

where $g_B(\ell)$ is g_B shifted by ℓ in time, the bar denotes removal of the time mean, and β is learned. The aligned map is $\tilde{g}_B = \sum_\ell w_\ell\, g_B(\ell)$; this is a differentiable estimate of the residual delay that the network can sharpen or soften.

Fusion and head. The fused map $[\,|g_A - \tilde{g}_B|,\ g_A \odot \tilde{g}_B\,]$ passes two convolution blocks and mean and max pooling; an MLP receives the pooled features together with ρ and outputs

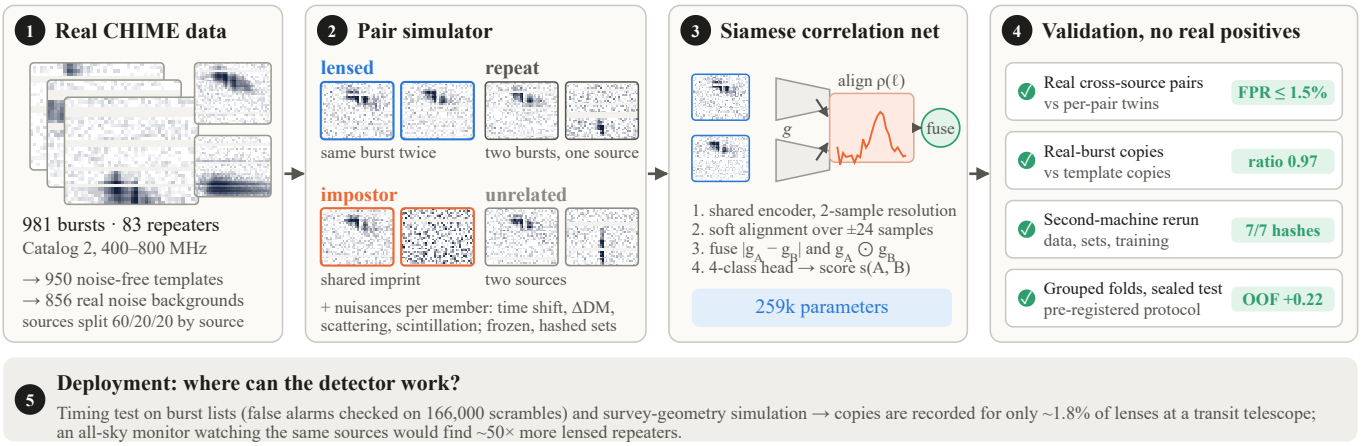


Fig. 2. Workflow. (1) Real CHIME/FRB Catalog 2 dynamic spectra (tiles: real bright bursts) provide noise-free templates and real noise backgrounds. (2) The pair simulator places templates on different real backgrounds with independent nuisances to make four classes; impostors share a spectral envelope or scintillation pattern. (3) The Siamese correlation network scores each pair. (4) Validation uses only real negatives, a real-burst proxy and an independent rerun. (5) Survey-geometry analysis decides where the detector can be deployed.

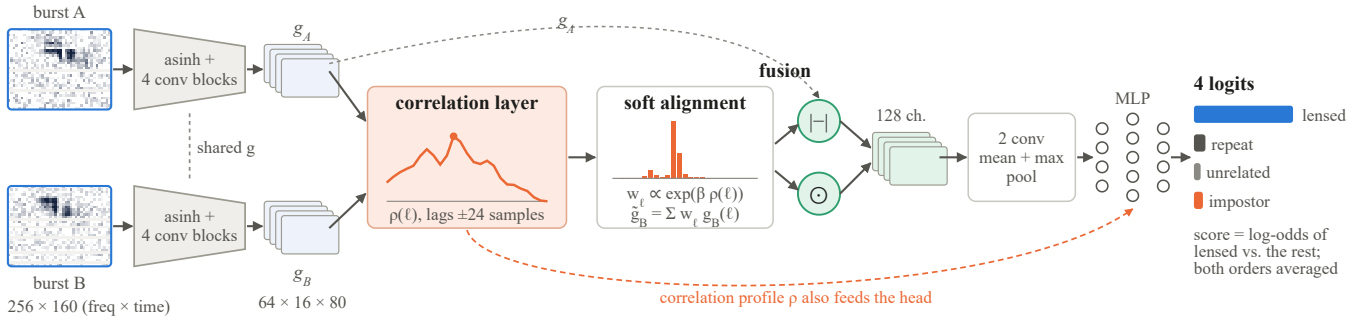


Fig. 3. Model 6. A weight-shared encoder maps each burst to a $64 \times 16 \times 80$ feature map at two-sample time resolution. The correlation layer measures the normalised correlation $\rho(\ell)$ over lags of ± 24 samples (curve: computed for the pair shown), softmax weights align g_B to g_A, and the aligned maps are fused by absolute difference and product. The correlation profile also feeds the head. Both directions are computed and averaged.

four logits z. The lensed score is the log-odds against the other classes, and symmetry is enforced by averaging both directions:

$$s(A,B) = \tfrac{1}{2}\left[\sigma(A,B) + \sigma(B,A)\right], \quad \sigma = z_{\mathrm{L}} - \log \sum_{c \neq \mathrm{L}} e^{z_c}. \tag{3}$$

The model has 259k parameters.

Training. All learned models are trained for 20 000 steps of 64 pairs drawn on the fly from training sources (half with S/N floor 10, half 25), with four-class cross-entropy, AdamW (learning rate 10^{-3}, weight decay 10^{-4}), 1000 warm-up steps and cosine decay; the checkpoint is selected on the validation bright set.

VI. Validation Without Real Positives

Grouped folds. Because the validation interval was wide, a pre-registered rule moved the primary comparison to five source-grouped folds, in which every source is scored by a model that never saw it. Here we report the rehearsal of this protocol on the 60 development sources (training and validation; test sources excluded), with one model-6 seed per fold; the final folds over all 80 sources follow the test sign-off.

True-negative transfer. Two bursts from different sources cannot be lensed copies, so real cross-source pairs are labelled negatives that can be scored without breaking the blind protocol. For each real pair we simulate five *twins* from the same two templates at their measured S/N and set τ at the twins' 99th percentile; the pre-registered criterion is a real false-positive rate of at most 2% per S/N bin. Twin matching is essential: with a threshold from generic simulated pairs, cross-correlation appears to fail below S/N 25 (real rate 6.2, 3.8, 2.4% for three variants), yet its twins fail as often (5.6, 4.0, 2.5%) because those real pairs are brighter than generic ones (median S/N 16.9 vs 13.5). The apparent failure is sample composition, not a simulation gap.

Real-burst positives. As a proxy for lensed pairs, two copies of a real burst's cutout, with its fine structure and real noise, are injected with independent nuisances on different backgrounds, alongside template copies and noise-only copies drawn with the same random numbers.

Reproduction. A second machine re-downloaded and re-

TABLE I

RECALL AT 1% FPR (LENSED VS REPEAT + IMPOSTOR), S/N $\geq$ 25; FIXED SPLIT: VALIDATION SOURCES; GROUPED FOLDS: OUT-OF-FOLD OVER THE 60 DEVELOPMENT SOURCES; 95% SOURCE BOOTSTRAP.

Method	Fixed split (3 seeds)	Grouped folds (OOF)	AUC
(1) Cross-correlation	0.171 [0.115, 0.245]	–	0.885
(2) Physics fit	0.164 [0.119, 0.235]	–	0.931
(3) Feature GBM	0.285 [0.180, 0.401]	0.246 [0.199, 0.297]	0.925
(4) Two-channel CNN	0.423 / 0.409 / 0.407	–	0.959
(5) Classic Siamese	0.500 / 0.559 / 0.541	–	0.976
(7) Cross-attention	0.471 / 0.013[a] / 0.579	–	0.976
(6) Ours	**0.701 / 0.698 / 0.684**	**0.467 [0.253, 0.662]**	**0.987**
(6) − (3), paired	+0.394 [+0.226, +0.542]	+0.220 [+0.035, +0.381]	
(6), second machine[b]	0.668 / 0.649 / 0.616	–	0.984–0.987

[a]One of three seeds collapsed to a plateau. [b]Different data-loader worker count; seed 0 inside the pre-registered range 0.66–0.72, the mean 0.05 below the original seeds. AUC: mean over seeds.

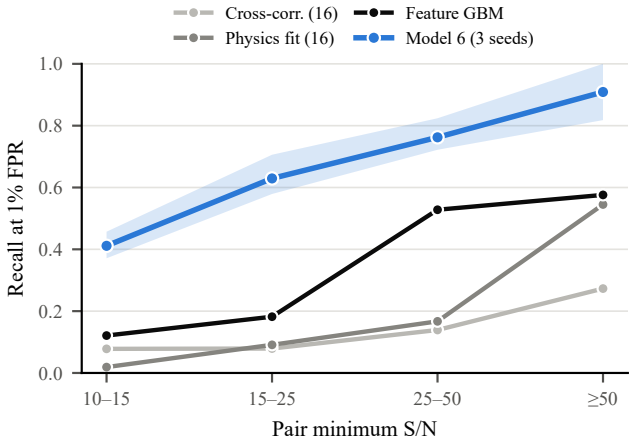


Fig. 4. Recall at 1% FPR by pair minimum S/N on the frozen validation set (529, 330, 108 and 33 lensed pairs per bin). Model 6: mean and range of three seeds from the second machine, checkpoints selected on the separate bright set.

processed all 981 spectra, rebuilt the injection cache, regenerated the frozen sets, retrained model 6 with three seeds and reran the checks.

VII. EXPERIMENTS

Baselines. All methods score the same pairs, averaged over both member orders: (1) normalised cross-correlation in the on-pulse box, maximised over ± 24 lags, with 16 channel blocks; (2) a physics statistic that fits one member to the other over scattering time, residual dispersion, lag and amplitude; (3) gradient-boosted trees (GBM) on seven measured burst properties per member, the best of 1–3 and the pre-registered comparator; (4) a two-channel CNN; (5) a classic Siamese network comparing embedding vectors; and (7) cross-attention between time–frequency patches.

Main result. Every learned model beats the physics and feature baselines, and model 6 leads all of them (Table I): +0.39 recall over the GBM on the fixed split and +0.22 [+0.04, +0.38] in the grouped-fold rehearsal, where every source is unseen; recall varies by fold (0.12–0.73), and the lowest fold is set by one source with two bursts. A gap of this size at near-equal AUC (0.976–0.987 for the learned models) shows that the gain lies in the low-false-positive tail that a search uses.

Sensitivity and errors. The gain holds at every S/N and is largest in relative terms at the faint end, where model 6 recovers 0.41 of lensed pairs against 0.12 for the GBM (Fig. 4). At the 1% threshold, 35–38 of the 40 false positives of each second-machine seed are impostors with a shared envelope (I1 or I1+I2), at most one is a scintillation-only impostor and 2–4 are plain repeats: the remaining confusion is the physically hardest case by design. Model 6 classifies 95% of lensed pairs of unseen sources correctly, while its confusion of repeat and unrelated pairs on unseen sources (32% correct against 86% on seen ones) shows that part of its capacity memorises source identity, which does not affect the lensed score.

Ablations. Removing the alignment, or training without impostors, lowers recall in every seed (Table II); without impostors the network mistakes envelope-sharing bursts of one source for copies. Oversampling template-similar repeats helps slightly in every seed, but the interval includes zero, and on a closed-form benchmark this scheme biased a learned likelihood ratio low by about 1 dex, so we do not adopt it without a correction. Two training changes aimed at the source memorisation, merging the unrelated label into repeat and a random smooth bandpass per member, did not raise recall on unseen sources (paired differences -0.036 to -0.006, all intervals including zero; original machine, three seeds each).

TABLE II

ABLATIONS OF MODEL 6 ON THE SECOND MACHINE, THREE SEEDS EACH: SEED-MEAN RECALL AT 1% FPR AND PAIRED DIFFERENCE FROM THE FULL MODEL (VALIDATION BRIGHT SET, 95% SOURCE BOOTSTRAP).

Variant	Recall	Difference
Full model	0.644	–
No alignment (correlation off)	0.557	−0.087 [−0.186, −0.025]
No impostors in training	0.474	−0.170 [−0.254, −0.052]
Hard-negative repeats	0.698	+0.053 [−0.011, +0.082]

Real-data validation. On the 3851 real cross-source pairs of the validation sources, all 12 trained models and 3 baselines meet the twin-threshold criterion once short real windows are filled with off-pulse noise like the simulated backgrounds, and model 6's real-to-twin rank correlation reaches the twin-to-twin ceiling on both machines (Table III). Before the fill, model 6 flagged real pairs whose windows shared the same zero-padded edges; the twin test exposed this format mismatch, which the fill removes. Two copies of a real burst, with its fine structure, are recognised as often as two copies of its smooth template, and noise-only copies never are, so model 6 does not rely on template smoothness or shared noise; the classic Siamese network does (ratio 0.72–0.84).

Efficiency. Model 6 scores 1149 pairs per second in both orders on one RTX 5070 Ti (bfloat16), so all 78 701 same-source pairs of Catalog 2 take about 70 s.

TABLE III
REAL-DATA VALIDATION OF MODEL 6 ON TWO MACHINES (VALIDATION SOURCES; 95% SOURCE BOOTSTRAP WHERE GIVEN).

Check (criterion)	Original machine	Second machine
Real FPR at twin threshold, S/N ≥ 25 ($\leq 2\%$)	1.1 / 1.4 / 0.8%	1.5 / 1.3 / 1.0%
GBM, same check	0.9%	1.0%
Real–twin rank corr. / twin ceiling	0.91–0.94 / 0.88–0.91	0.91–0.92 / 0.87–0.89
Real-burst / template recall, S/N 25 (≥ 0.8)	0.96 [0.89, 1.06]	0.97 [0.91, 1.08]
Real-burst / template recall, S/N 40	0.99 [0.91, 1.13]	1.04 [0.97, 1.18]
Noise-only copies flagged (≤ 0.05)	0.000	0.000
Frozen-set hashes	reference	7/7 match

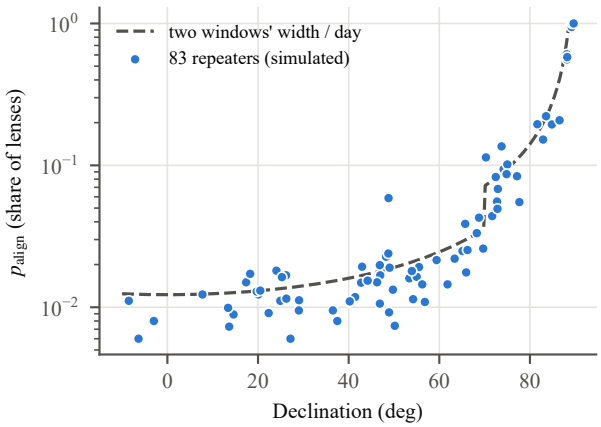


Fig. 5. Share of lens delays for which CHIME records the copies of a repeater's bursts: the 83 Catalog 2 repeaters (simulated with their own burst times, exposure and redshift) and (4) (dashed).

VIII. SURVEY GEOMETRY: WHERE CAN THE DETECTOR WORK?

A lensed repeater can also be found without a classifier. A galaxy lens fixes Δt to far better than the 4 ms catalogue timing, so two burst pairs (i, j), (k, l) of one source with $|\Delta t_{ij} - \Delta t_{kl}|$ below four combined timing errors already form a significant pattern. We calibrated this timing test on burst lists scrambled by whole sidereal days, which keeps each burst's hour angle, and confirmed its false-alarm rate on independent scrambles (1 of 166,000 held-out scrambles).

Both the classifier and the timing test need the copies to be recorded. CHIME is a transit telescope: every recorded burst arrives within a window of width $W(\delta) = 2\theta_{\rm EW}/(15^\circ\,{\rm h}^{-1}\cos\delta)$ around transit, with $\theta_{\rm EW} \approx 1.1^\circ$ measured from the real bursts. The copy of each burst arrives at the same hour-angle offset, $24\,{\rm h} \times {\rm frac}(\Delta t/T_{\rm sid})$, so copies are recorded for all bursts or none (Fig. 1b), and for a broad delay distribution

$$p_{\rm align} \approx 2W(\delta)/T_{\rm sid}, \tag{4}$$

doubled above $\delta = 70^\circ$, where lower transits add a window. Simulating singular-isothermal-sphere lenses with each source's redshift, burst times and exposure gives a median $p_{\rm align}$ of 1.8%, following (4) (Fig. 5). Repeating FRBs therefore do not escape the transit problem known for one-off bursts [22].

TABLE IV
EXPECTED NUMBERS OF LENSED REPEATERS AMONG THE 83 CATALOG 2 SOURCES (γ: MAGNIFICATION-BIAS SLOPE).

Survey	CHIME, timing	CHIME, ≥ 1 copy	All-sky, timing
Catalog 2, as observed	5.3×10^{-6}	3.2×10^{-5}	2.6×10^{-4}
Catalog 2, $\gamma = 1.5$	1.2×10^{-4}	4.9×10^{-4}	6.1×10^{-3}
10-yr survey	7.4×10^{-6}	4.9×10^{-5}	3.7×10^{-4}
10-yr survey, $\gamma = 1.5$	2.2×10^{-4}	6.3×10^{-4}	7.4×10^{-3}

Table IV gives the expected number of detectable lensed repeaters. The Catalog 2 sample holds 5.3×10^{-6}–1.2×10^{-4}, so a null search is the expected outcome; a 10-year transit survey gains only 1.4–1.8$\times$ because the alignment requirement remains, while an all-sky monitor watching the same sources gains $\sim$50$\times$. The result changes by -19% when the median host-galaxy dispersion measure is raised from 68 to 150 pc cm^{-3}. For transit surveys, including next-generation drift-scan arrays such as CHORD [23], the classifier's role is to vet the rare single recorded copy and reject impostors; its full value lies with wide-field monitors such as BURSTT [24].

IX. DISCUSSION AND LIMITATIONS

Simulated false-positive rates rank methods, but deployment thresholds must come from real negatives, and twin matching is what makes such a check interpretable: it separates simulation gaps from differences in sample composition, and it exposed a real-versus-simulated format mismatch that no simulated benchmark would show. The limitations are equally concrete. The results are development results, because the test split stays sealed under the pre-registered protocol, and the grouped-fold numbers are a rehearsal on development sources. Recall falls with differential scattering between the two images: on validation pairs it halves at $\tau_{600} \approx 1$ ms and is 0.13 of its unscattered value at 3 ms, and training to 10 ms does not recover it, whereas the physics fit, which models scattering, keeps its recall to 3 ms. The detector therefore suits lenses with little cold gas, which dominate the lensing optical depth. The second machine reproduced the data bit for bit but placed model 6 about 0.05 recall below the original seeds, for reasons not yet isolated. Lenses are singular isothermal spheres, impostor strength rests on a simulation choice, and the validation set holds 26 sources.

X. CONCLUSION

A small Siamese network that aligns before it compares identifies lensed copies of real FRB spectra far better than physics and feature baselines within the simulated nuisance ranges, passes real-data checks designed for a detector without real positives, and reproduces on a second machine. The same analysis shows that transit geometry, not classification accuracy, is the bottleneck for finding lensed repeaters, which points such detectors to wide-field radio arrays. All inputs are public (CHIME/FRB Catalog 2, DOI 10.11570/25.0066); code, frozen-set manifests, hashes and pre-registered plans are in the project repository (link withheld for review).

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